Machine Learning-Based Prediction of Seropositive *Toxoplasma gondii* from Lifestyle Factors

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***Abstract*—**

***Toxoplasma gondii* is a parasitic infection that poses a serious threat to immunocompromised individuals, particularly people living with HIV, due to its often silent and undetected progression. Traditional serological testing can be costly and inaccessible in certain communities, resulting in many individuals going untested. To address this challenge, this study explores the use of machine learning (ML) models to develop an early screening tool based solely on questionnaire responses, offering a more accessible and cost-effective alternative to laboratory testing.**

The dataset used in this study contains behavioral, dietary, environmental, and socio-economic factors collected from individuals who had undergone IgG serological testing. After cleaning and encoding the data, feature engineering was performed to derive key composite features such as economic and dietary risk scores. Preprocessing steps included standardization, Principal Component Analysis (PCA), and class balancing using SMOTE.

Five supervised ML models—Random Forest, Logistic Regression, XGBoost, MLP, and SVM—were trained and optimized using GridSearchCV. The models were evaluated using weighted accuracy and F1-score. Performance varied across economic and dietary subgroups, with better results observed in data with higher economic scores and lower dietary risk. The Multi-Layer Perceptron (MLP) model achieved the highest performance, reaching an F1-score of 0.80 in this group. This highlights the relevance of the selected features and the influence of lifestyle on infection risk. Despite the limited sample size, the models showed promising potential for supporting early screening, especially in underserved populations.

***Keywords* — Toxoplasma gondii, HIV, early screening, machine learning, classification, feature engineering, GridSearchCV.**

I. Introduction

Toxoplasma gondii is a parasitic infection that is often asymptomatic in healthy individuals but can pose significant risks to immunocompromised individuals, particularly those with Human Immunodeficiency Virus (HIV). In HIV patients, the infection can lead to severe complications such as encephalitis, and it remains a leading cause of death among this vulnerable population [1]. Despite its potential dangers, routine testing for *Toxoplasma gondii* is rarely conducted due to the high costs and difficulties associated with diagnostic procedures, leading to a delay in treatment and higher mortality rates.

In light of these challenges, there is an urgent need for a more accessible and cost-effective method for screening *Toxoplasma* infection. An early screening tool, based on a questionnaire that estimates the risk of infection, could provide HIV patients with a preliminary understanding of their health status, helping them to make informed decisions before seeking formal medical diagnosis. Such a tool would be particularly valuable in regions where access to healthcare is limited or expensive.

Previous research has shown a clear connection between HIV and an increased risk of *Toxoplasma* infection, with several factors contributing to the higher prevalence in this population. Meirawan et al. (2023) [2] examined the role of sexual behavior and the history of injecting drug use as significant predictors of *Toxoplasma* infection in HIV patients. Other studies have suggested that immune suppression, a hallmark of HIV, exacerbates susceptibility to various infections, including *Toxoplasma*. Moreover, research has demonstrated that the cost of diagnostic tests for *Toxoplasma* remains a significant barrier to early detection, further highlighting the importance of developing more affordable and accessible screening options.

Recent studies have also explored the use of machine learning for *Toxoplasma* diagnosis. Alvarez et al. [3] developed an AI-based model to detect ocular toxoplasmosis using quantization and pruning techniques. Cifuentes et al. [4] applied AutoML to classify ocular toxoplasmosis from fundus

images, achieving high diagnostic performance. These findings support the growing relevance of machine learning as a non-invasive and scalable diagnostic approach—motivating this study's questionnaire-driven prediction model.

This paper presents the development of a machine learning-based model for early screening of *Toxoplasma gondii* infection in HIV patients, aiming to provide a cost-effective and easily accessible method for estimating infection risk. By leveraging questionnaire responses, this tool will allow individuals to assess their potential need for further medical diagnosis, ultimately improving healthcare accessibility and early intervention for this at-risk group.

II. Related Works

Meirawan and Ramadhani (2023) [2] investigated the correlation between sexual behaviors and the risk of *Toxoplasma gondii* infection in HIV-positive homosexual men. Their study identified that certain sexual behaviors, such as receiving payment for sex and engaging in unprotected oral sex, were significantly associated with higher seropositivity for *T. gondii*.

The relationship between hygiene, environment, and toxoplasmosis has been explored in other contexts as well. Marthalia et al. [5] found a high prevalence of chronic toxoplasmosis in cat breeders, with poor personal and environmental hygiene being major risk factors. Similarly, Athiyyah et al. [6] showed that poor sanitation and hygiene practices were strongly associated with intestinal parasitic infections among children in Indonesia, underscoring the significance of behavioral and environmental features, which are also used in our model.

Sami et al. [7] highlighted the risk of transmission through poultry consumption, identifying *T. gondii* Genotype I in chickens. Ducrocq et al. [8] further confirmed the increased risk of infection from eating raw or undercooked meat through a meta-analysis, reinforcing the role of dietary habits in toxoplasmosis transmission. These findings justify the inclusion of food consumption behavior as a predictive factor in our model.

Khairullah et al. [9] provided a broad review of toxoplasmosis, outlining key transmission routes, including contact with cat feces, contaminated food and water, and congenital infection, many of which are captured in the questionnaire used for this study. Stelzer et al. [10] emphasized the significance of livestock in transmission and associated public health risks, while also noting the economic impact of toxoplasmosis on agriculture.

Azizi et al. [11] demonstrated the prevalence of T. gondii genotype III in stray cats in Iran, reinforcing the importance of cats as a primary source of environmental contamination. Similarly, Mosquera et al. [12] showed how diet and habitat exposure influenced seropositivity rates in Galapagos birds, confirming trophic and environmental transmission pathways.

Recent studies have also focused on seroprevalence in specific populations. Butt et al. [13] conducted a meta-analysis on blood donors, revealing that residence area and contact with cats significantly affected seroprevalence rates. Sharifzadeh et al. [14] found a higher rate of *T. gondii* infection among drug-addicted individuals, suggesting that lifestyle-related vulnerability can greatly influence infection risk.

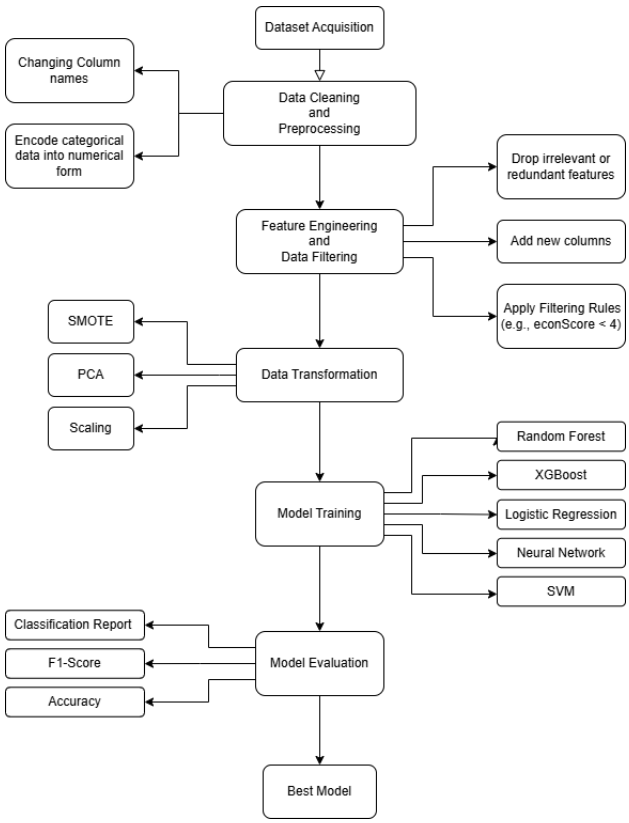


Fig. 1: Research Flow Diagram

The application of AI-based solutions for toxoplasmosis diagnosis has gained attention. Alvarez et al. [3] created an efficient diagnostic model using TensorFlow Lite, achieving excellent F1-scores while reducing computational cost. Cifuentes et al. [4] leveraged cloud-based AutoML platforms to detect and classify ocular toxoplasmosis from fundus images with strong diagnostic accuracy, showcasing the value of ML in infectious disease prediction.

These studies provide a foundation for utilizing behavioral, environmental, and clinical data to predict *Toxoplasma gondii* seropositivity. However, there is limited research combining such data with machine learning models for early screening in resource-limited settings, a gap this thesis aims to address.

III. Research Methodology

A. Research Flow

We began our research by exploring the dataset used for this study, which contains information related to seropositivity for *Toxoplasma gondii*. This dataset includes a range of lifestyle and health-related questionnaire responses, along with corresponding blood test results indicating seropositivity. These features were used as parameters for the machine learning models applied in our study. The overall research flow diagram is presented in Figure 1.

IV. Dataset

This study used data collected from individuals living with HIV undergoing diagnosis, treatment, and care. This population was chosen due to their heightened vulnerability to severe complications from *Toxoplasma gondii* infection.

The dataset consists of questionnaire responses covering lifestyle factors such as diet, pet contact, environmental exposure, and hygiene habits. It also includes `serostatus` results indicating antibody presence. In total, 186 records were collected, with 123 testing seropositive and 63 seronegative, showing a class imbalance (see Table I).

TABLE I: Dataset

Variable		Amount	Prevalence
Serology Status igG	positive	123	66.13%
Toxoplasma gondii	negative	63	33.87%

The dataset was cleaned and preprocessed by renaming columns, encoding categorical values, and removing corrupted or irrelevant entries.

TABLE II: Dataset Columns

Variable Name	**Description**
Titer	IgG antibody level
Serostatus	1 = positive, 0 = negative
Education	Scale 1–7, higher values indicate less education
Condom Use During Oral Sex	Scale 0–4, higher values indicate less frequent use
Partner's Condom Use (Oral)	Scale 0–4, higher values indicate less frequent partner use
Frequency of Anal Intercourse	Scale 0–4, higher values indicate greater frequency
Current Injection Drug Use	1 = Yes, 0 = No
History of Used Needle Use	Scale 0–3, with higher values indicating more frequent use
Nail Cutting Routine	Scale 0–4, with higher values indicating lower nail maintenance
Grilled/Smoked Poultry Consumption	Scale 0–4, indicating frequency of consumption
Soil Contact Frequency	Scale 0–4, indicating frequency of contact with soil
Cat Feces Exposure	Scale 0–4, indicating frequency of observing cat feces near residence
Glove Use During Soil Contact	Scale 0–4, where higher values indicate less frequent glove use
Tooth Loss Due to Extraction	Numeric Count
Tooth Loss Due to Natural Causes	Numeric Count
Mouth Ulcers (Last 3 Months)	Frequency of experiencing mouth ulcers over the past 3 months
Persistent Mouth Ulcers (1+ Month)	Frequency of ulcers lasting more than one month
Economic Score	Wealth score based on assets
Hygiene Score	Hygiene behavior score
Sexual Risk Score	Sexual risk behavior score
Condom Use Score	Condom use behavior score
Miscellaneous Risk Score	Other risk factor score (Drug use, injury during intercourse)

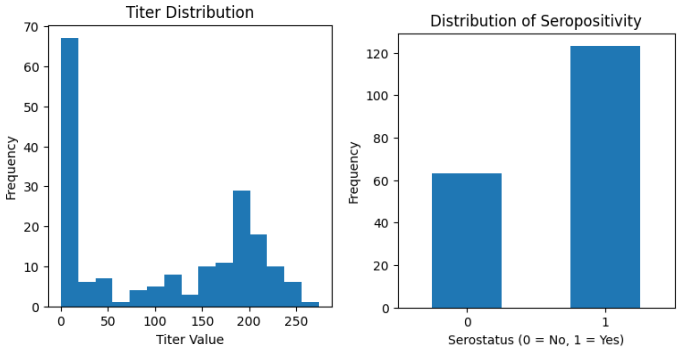
Feature engineering followed, where new variables such as Economic Score and Dietary Risk Score were derived to reflect socioeconomic status and dietary behavior. The Economic Score captured ownership of assets like vehicles and internet access, while the Dietary Risk Score reflected frequency of consuming raw or high-risk foods.

Feature filtering involved expert consultation, boxplot visualization, SHAP analysis, and correlation checks. Features with low relevance (correlation < 0.05) or unclear patterns were removed to reduce noise.

Row-level filtering excluded entries with incomplete scores or extreme outliers to refine the dataset further. After filtering, 78 entries remained 55 seropositive and 23 seronegative indicating stronger imbalance.

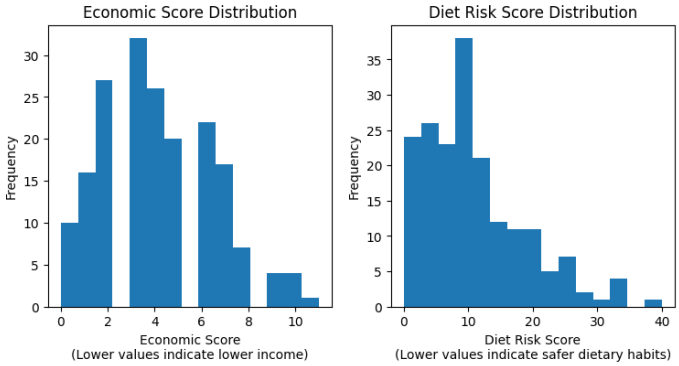
The data were then split into training and testing sets (70/30), resulting in a test set with 5 seronegative and 19 seropositive samples. This class imbalance further justified using weighted evaluation metrics.

Final preprocessing included standardization, dimensionality reduction using PCA, and class balancing with SMOTE. These steps ensured the dataset was suitable for training machine learning classifiers to predict seropositivity based on questionnaire responses.



(a) (b)

Fig. 2: Graph of titer distribution and seropositivity



(a) (b)

Fig. 3: Graph of economic score and dietary risk score distribution

A. Preprocessing

1) Preprocessing Flow:

- **Column Renaming:** The original dataset consisted of very long questionnaire-based column names. These were

renamed to concise, readable feature names for ease of use and analysis.

- **Numerical Encoding:** All categorical data such as gender, education, and various behavioral attributes were converted to numerical format to allow compatibility with machine learning algorithms.
- **Feature Engineering:** We created new features to better represent aggregated behaviors and risk categories based on the existing questionnaire responses. These included features like `age`, `economicScore`, `sexScore`, and many others.
- **Feature Selection:** Using insights from domain experts, boxplots were reviewed to evaluate outliers and distributions. Columns deemed irrelevant or noisy were dropped from the dataset.
- **Correlation Filtering:** Features with weak correlation to the target variable (`serostatus`) were removed to prevent overfitting and improve model performance.
- **Principal Component Analysis (PCA):** PCA was used to reduce dimensionality while retaining important variance in the dataset, improving model efficiency and generalizability.
- **SMOTE:** As the dataset was imbalanced, the Synthetic Minority Oversampling Technique (SMOTE) was applied to generate synthetic samples of the minority class and balance the dataset.

B. Machine Learning Techniques and Algorithms

This study adopted a supervised learning approach to predict seropositive *Toxoplasma gondii* infections based on questionnaire-derived features. The objective was to develop a model that can estimate the risk of seropositivity without relying on costly serological tests, thereby enabling early screening, especially for HIV-positive individuals. The following machine learning algorithms were applied and evaluated:

1) **Logistic Regression (LR)**
   A statistical modeling technique used for binary classification that estimates the probability of a dichotomous outcome based on one or more independent variables. It uses the logistic function to map predicted values to probabilities between 0 and 1 [15].
2) **Support Vector Machine (SVM)**
   A supervised learning algorithm that constructs an optimal separating hyperplane between classes by maximizing the margin between support vectors. Its effectiveness is enhanced by the kernel trick, which allows it to handle nonlinear decision boundaries [16].
3) **Random Forest (RF)**
   A robust and user-friendly ensemble learning algorithm that constructs multiple decision trees using randomized subsets of data and features, and combines their outputs to enhance prediction accuracy and minimize overfitting. This method is effective for both classification and regression tasks and is particularly valued for its ability to handle high-dimensional data and assess feature importance [17].
4) **XGBoost (Extreme Gradient Boosting)**
   A highly efficient and scalable implementation of gradient boosting that incorporates regularization, sparsity awareness, and parallel processing to optimize performance on large datasets [18].
5) **Multilayer Perceptron (MLP)**
   A Multilayer Perceptron (MLP) is a widely used type of feedforward artificial neural network composed of an input layer, one or more hidden layers, and an output layer, where each layer consists of interconnected processing elements known as neurons. This architecture enables MLPs to model complex, nonlinear relationships between input features and target outputs [19].

The raw dataset underwent a series of preprocessing steps, including feature renaming, feature selection based on expert-guided analysis, and encoding. To address class imbalance, the Synthetic Minority Oversampling Technique (SMOTE) was applied. Feature scaling was performed using the StandardScaler, and dimensionality reduction was conducted using Principal Component Analysis (PCA), retaining components that explained 95% of the variance.

Hyperparameter tuning was conducted using GridSearchCV, a method that systematically tests a range of parameter combinations to identify the configuration that yields the best cross-validated performance for each model. Final model performance was evaluated using accuracy and F1-score.

C. Machine Learning Pipeline

Input Dimension: 20 features $\rightarrow$ n PCA components (retaining 95% variance).
Output Dimension: Binary (seropositive: 1, seronegative: 0).

Model Architectures:

- **Random Forest**: 100 trees, max depth=5, Gini impurity splitting.
- **XGBoost**: 100 trees, max depth=7, learning rate=0.01.
- **MLP**: 1 hidden layer (50 neurons, tanh activation), Adam optimizer.
- **Logistic Regression**: Balanced class weight, l1 ratio=0.3, C=0.01.
- **SVM**: C=10, gamma=scale, kernel=rbf

D. Evaluation Method

To evaluate the performance of the classification models, we used two standard metrics: **accuracy** and **F1-score**. These metrics were selected after considering the class imbalance present in the dataset, where 66.13% of the instances are seropositive and 33.87% are seronegative.

1) **Accuracy**
   Accuracy measures the proportion of correct predictions across all classes. Although it can be misleading in imbalanced datasets since it may favor the majority class it is included in this study for completeness [20]. Accuracy is calculated as:

$$\text{Accuracy} = \frac{\text{True Positives} + \text{True Negatives}}{\text{Total Predictions}}$$

2) **F1-Score**
The F1-score is the harmonic mean of precision and recall, providing a balanced evaluation of the model's performance. It is particularly suitable for imbalanced datasets, as it penalizes models that perform well on the majority class but poorly on the minority class [20]. The F1-score is defined as:

$$\text{F1} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Because of the class imbalance, both metrics were computed using **weighted averages**, which account for the proportion of each class in the dataset. This ensures that performance across all classes is fairly represented in the final evaluation.

V. Results and Analysis

TABLE III: Model Evaluation
(Economic Score > 4 and Dietary Risk Score < 18)

Algorithm	accuracy	f1-score
RandomForest	0.76	0.77
LogisticRegression	0.75	0.76
XGBoost	0.7	0.72
MLP	0.78	0.8
SVM	0.77	0.79

TABLE IV: Model Evaluation
(Economic Score < 4 and Dietary Risk Score > 18)

Algorithm	accuracy	f1-score
RandomForest	0.46	0.47
LogisticRegression	0.53	0.54
XGBoost	0.52	0.51
MLP	0.44	0.45
SVM	0.49	0.5

In this study, the evaluation metrics for classification performance are reported using weighted averages for accuracy and F1-score. This approach takes into account the class imbalance present in the dataset by weighing each class's metric by its respective number of instances. As the positive class significantly outnumbers the negative class, using weighted averages ensures a more balanced and representative assessment of the model's performance across both classes.

From Table III, which evaluates models trained on samples with an Economic Score above 4 and a Dietary Risk Score below 18, we observe relatively high and consistent model performance. The Multi-Layer Perceptron (MLP) and Support Vector Machine (SVM) classifiers achieve the highest F1-scores of 0.80 and 0.79, respectively, while other models such as Random Forest and Logistic Regression also perform well, maintaining F1-scores above 0.75. This consistency across multiple models suggests that the features used in this study are effective in capturing the patterns related to seropositivity within this subgroup.

In contrast, Table IV shows a significant decline in performance for the same models when trained on data where the Economic Score is below 4 and the Dietary Risk Score exceeds 18. F1-scores drop to a range between 0.45 and 0.54, reflecting a reduced ability of the models to make accurate predictions in this context. Despite the decline, the fact that the models are still able to make moderately effective predictions under these conditions further supports the relevance of the selected features, even when applied to more challenging data segments.

The performance gap between the two scenarios highlights the relevance and influence of the chosen features — particularly those related to economic conditions and dietary behavior — in the prediction of Toxoplasma gondii seropositivity. The fact that these features contribute to strong model performance in the first scenario, and still retain some predictive value in the second, emphasizes their importance in understanding and modeling the infection risk. These findings support the notion that socioeconomic and dietary lifestyle factors play a significant role in exposure risk and should be considered as key indicators in early screening models.

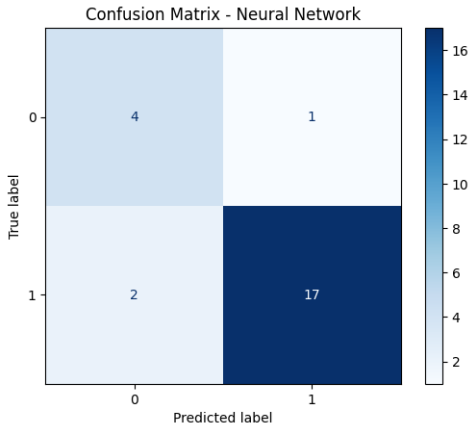


Fig. 4: MLP Confusion Matrix

VI. Conclusion and Discussion

This study applied machine learning to predict *Toxoplasma gondii* seropositivity using questionnaire-derived lifestyle and socioeconomic data. The engineered composite features—particularly the `economicScore` and `dietaryRiskScore`—proved highly relevant for risk prediction, with models achieving strong performance (F1-scores 0.76–0.80) in populations with favorable economic conditions (Economic Score >4) and lower dietary risk (Dietary Risk Score <18). These findings align with known transmission pathways, as individuals with lower economic status typically face greater exposure to potential infection sources like contaminated soil and undercooked meat.

Among the evaluated algorithms, the Multi-Layer Perceptron (MLP) demonstrated the highest predictive accuracy, followed closely by Support Vector Machines (SVM), suggesting that non-linear relationships between risk factors and seropositivity may be important. However, model performance declined significantly (F1-scores 0.45–0.54) when applied to disadvantaged populations (Economic Score <4 and Dietary

Risk Score >18), indicating either that additional unmeasured risk factors are present in these groups or that different predictive models may be needed for different socioeconomic strata.

The study's primary limitation is the relatively small dataset size (N=78), which constrains model generalizability despite the use of SMOTE oversampling and rigorous hyperparameter tuning. Furthermore, the binary serostatus classification does not capture potentially useful information about infection timing or antibody titers that could aid in more nuanced risk stratification.

To address the challenge of limited data, future research may benefit from exploring synthetic data generation techniques beyond SMOTE, such as Conditional Tabular GANs (CTGAN), which can model complex feature distributions and generate realistic new samples. [21].

These results suggest several promising directions for future work. Expanding the dataset to include more granular behavioral data—such as food preparation methods and pet care practices—and validating models on larger, more demographically varied populations could enhance model accuracy and generalizability. The demonstrated feasibility of this questionnaire-based approach supports its potential as a low-cost, accessible screening tool, particularly in resource-limited settings where serological testing is unavailable. This work establishes a strong foundation for developing more scalable and equitable risk assessment methods for toxoplasmosis.

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